

BEATRICE, NE, USA

WMO: 725515

Lat: 40.301N

Lon: 96.754W

Elev: 404

StdP: 96.57

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 94947

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.2	-16.0	-22.9	0.5	-17.6	-20.2	0.7	-14.3	14.8	-2.1	12.9	-4.0	3.3	320	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	36.0	23.6	34.0	23.3	32.5	23.0	25.6	32.3	24.9	31.4	24.1	30.2	5.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.8	19.6	29.3	22.9	18.5	28.4	22.3	17.9	28.0	81.2	32.4	77.8	31.4	74.7	30.3	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.9	9.3	DB	-23.3	38.5	2.6	1.5	-25.2	39.6	-26.7	40.5	-28.2	41.4	-30.1	42.5	(4)
(5)			WB	-23.7	27.1	2.5	0.9	-25.5	27.8	-26.9	28.3	-28.3	28.8	-30.2	29.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	-3.2	-0.8	5.3	11.3	17.3	23.0	25.5	24.1	19.7	12.5	5.0	-1.4	(6)	
	DBStd	11.35	6.61	6.74	6.54	5.42	4.83	3.74	3.53	3.32	4.74	5.34	5.66	6.02	(7)	
	HDD10.0	1502	409	307	174	48	3	0	0	0	1	35	168	356	(8)	
	HDD18.3	3162	667	537	406	218	78	6	1	1	41	193	402	613	(9)	
	CDD10.0	2077	0	3	28	87	231	389	481	437	292	112	17	1	(10)	
	CDD18.3	697	0	0	1	6	47	146	224	180	81	11	0	0	(11)	
	CDH23.3	7391	0	0	18	98	502	1517	2518	1780	831	123	4	0	(12)	
(13)	CDH26.7	3144	0	0	3	23	169	621	1207	774	317	30	1	0	(13)	
(14)	Wind	WSAvg	4.2	4.2	4.4	4.8	5.2	4.6	4.1	3.4	3.2	3.7	4.2	4.2	4.1	(14)
(15) Precipitation	PrecAvg	771	18	21	54	68	114	106	101	95	85	57	35	20	(15)	
	PrecMax	1135	54	79	182	176	284	220	295	262	302	160	93	62	(16)	
	PrecMin	340	0	0	0	2	38	8	2	10	8	0	0	0	(17)	
	PrecStd	163	13	18	43	36	58	47	61	59	62	40	27	16	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.0	20.5	25.3	29.4	33.8	36.0	38.6	37.5	35.0	30.2	23.4	16.6	(19)	
		MCWB	7.5	10.9	15.3	17.3	21.0	23.1	23.5	23.1	22.4	19.4	13.9	10.8	(20)	
	2%	DB	11.8	16.0	22.4	26.2	30.9	33.8	36.6	35.0	32.6	26.7	19.9	12.7	(21)	
		MCWB	6.2	9.1	13.7	15.8	20.1	23.2	24.0	23.5	22.0	17.2	12.0	7.7	(22)	
	5%	DB	8.3	12.2	19.1	23.2	28.1	32.3	34.4	32.8	30.6	23.9	17.2	10.1	(23)	
		MCWB	4.3	7.3	12.3	14.7	18.9	23.0	23.9	23.4	21.2	16.2	11.2	6.0	(24)	
	10%	DB	5.9	8.7	16.1	21.0	26.2	30.3	32.6	31.1	27.9	22.0	14.1	7.4	(25)	
		MCWB	2.9	4.9	10.6	13.7	18.2	21.9	23.6	22.9	19.9	15.1	9.1	4.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.6	13.5	17.6	19.2	22.9	26.1	26.9	26.3	24.8	21.5	15.6	12.7	(27)	
		MCDB	14.1	17.5	22.7	25.7	30.2	32.5	33.4	31.9	31.3	26.7	19.5	15.2	(28)	
	2%	WB	6.6	9.7	15.6	17.5	21.7	24.9	25.8	25.2	23.6	19.5	13.8	8.6	(29)	
		MCDB	10.8	15.2	19.3	23.5	28.1	31.3	32.9	31.5	29.5	24.2	18.6	11.4	(30)	
	5%	WB	4.8	7.1	13.3	15.9	20.6	24.0	25.1	24.3	22.5	17.6	11.7	6.0	(31)	
		MCDB	8.5	11.5	18.2	21.7	26.4	30.2	32.0	30.5	28.0	21.9	16.2	9.7	(32)	
	10%	WB	2.9	5.0	10.7	14.3	19.2	22.9	24.3	23.6	21.4	15.7	9.7	4.1	(33)	
		MCDB	5.7	8.8	16.4	20.0	24.7	28.9	30.6	29.4	26.6	20.4	14.2	7.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	11.0	12.3	12.5	11.7	11.3	11.4	11.4	12.7	12.8	12.0	10.5	(35)	
		MCDBR	15.3	16.5	17.1	15.9	14.5	12.9	13.3	13.6	14.4	15.6	16.0	15.1	(36)	
		MCWBR	10.5	10.5	9.6	8.6	6.6	5.4	4.7	5.4	5.9	8.0	9.6	10.5	(37)	
	5% WB	MCDBR	14.6	15.0	13.8	14.1	12.2	11.5	11.5	11.8	11.9	12.4	14.4	13.7	(38)	
		MCWBR	10.4	10.1	8.6	8.6	6.3	5.8	5.2	5.5	5.8	7.5	9.6	10.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.281	0.297	0.323	0.372	0.404	0.414	0.421	0.423	0.380	0.334	0.295	0.281	(40)	
	taud		2.479	2.450	2.420	2.305	2.284	2.312	2.291	2.292	2.360	2.476	2.511	2.484	(41)	
	Ebn at Noon		893	925	932	900	875	864	853	842	858	869	868	863	(42)	
	Edh at Noon		76	91	104	126	131	128	130	126	110	88	73	70	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.18	3.04	4.00	5.05	5.90	6.63	6.58	5.79	4.75	3.42	2.43	1.85	(44)	
	RadStd		0.18	0.20	0.33	0.32	0.44	0.42	0.42	0.28	0.31	0.36	0.18	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+0.55	N/A	N/A	N/A	N/A
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page